

Deep Learning-Based Sign Language Gesture Recognition

MOHAMMAD SALMAN¹, (Member, UMT), MUAZ HASSAN², (Member, UMT)

¹Department of AI, UMT (University of Management and Technology), Lahore, Pakistan (e-mail: s.khizar.t@gmail.com)

²Department of AI, UMT (University of Management and Technology), Lahore, Pakistan (e-mail: muazhsanansari@gmail.com)

Corresponding author: Mohammad Salman (e-mail: s.khizar.t@gmail.com).

ABSTRACT This research investigates the application of Convolutional Neural Networks in recognizing sign language gestures. The deep learning architecture classifies gestures based on a dataset of static images representing hand signs. Through data augmentation techniques, it is hoped the generalization of the model improves and is examined in terms of metrics based on accuracy, loss, and confusion matrix evaluation. The work presents a new set of tools that actually aid communication for the hearing and speech-impaired persons.

Keywords Sign language recognition, deep learning, convolutional neural networks, assistive technology, Transfer learning, mobileNetV2, Pre-trained model.

I. INTRODUCTION

Communication barriers have often led to isolation for individuals relying on sign language. Automated systems which help interpret sign language will greatly enhance inclusivity and communication. This study will be successful in signifying one of the bridges towards breaking communication gaps by classifying static images of sign language gestures using Convolutional Neural Networks (CNNs).

Deep learning, especially CNNs, has found marking success in image classification problems. CNNs efficiently extract spatial features from images, endowing those with the ability to recognize hand gestures in sign language. In this work, a structured CNN model is applied to establish a deep and accurate model for gesture recognition that employs data augmentation techniques. Additionally, this work also uses transfer learning to apply MobileNet, a pre-trained model, in improving model performance further.

II. OBJECTIVES

The basic aims of this study are:

- Development of a CNN-based model to classify static images of sign language gestures;
- Research into the data augmentation effect on model performance and robustness;
- Evaluate model performance using relevant metrics and suggest improvement points;
- Transfer learning employing MobileNet, a pre-trained model, to enhance classification performance.

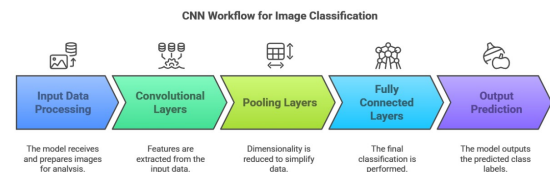


FIGURE 1: Workflow in CNN Model

The goals contribute to the bigger vision of providing an assistive tool in interpreting sign language and enhancing communication for individuals with hearing and speech impairments.

III. TRANSFER LEARNING

Transfer learning in this research involves the utilization of MobileNet, a pre-trained model, for higher accuracy rates in classification tasks. MobileNet operates efficiently with low power requirements, making it apt for real-time gesture recognition applications. Thanks to the use of a pre-trained model, the system incorporates knowledge gained through prior training on large datasets during the subsequent learning phases, thus minimizing data requirements and computational resources needed for training.

$$y = x + T(x) \quad (1)$$

where:

- x is the input tensor.
- $T(x)$ is the transformation function (convolutional layers).
- The residual connection is applied only if input and output dimensions match.

The transformation function $T(x)$ consists of three main steps:

A. 1. EXPANSION LAYER (1×1 CONVOLUTION + RELU6)

$$x' = \sigma(W_e x) \quad (2)$$

where:

- W_e is the expansion weight matrix.
- σ is the ReLU6 activation function.
- x' is the expanded feature map.

B. 2. DEPTHWISE CONVOLUTION (3×3 DEPTHWISE CONV + RELU6)

$$x'' = \sigma(W_d * x') \quad (3)$$

where:

- W_d represents the depthwise convolution filters.
- $*$ denotes the depthwise convolution operation.

C. 3. PROJECTION LAYER (1×1 CONVOLUTION, NO ACTIVATION)

$$y = W_p x'' \quad (4)$$

where:

- W_p is the projection weight matrix.
- The projection reduces dimensionality without activation.

D. FINAL BOTTLENECK RESIDUAL BLOCK REPRESENTATION

$$y = W_p \sigma(W_d \sigma(W_e x)) \quad (5)$$

For an input image X , MobileNetV2 extracts features as:

$$F = f_\theta(X) = \text{GlobalAveragePool}(W_p \sigma(W_d \sigma(W_e X))) \quad (6)$$

where:

- f_θ represents the MobileNetV2 feature extractor.
- F is the final extracted feature representation.

Using MobileNet, we can now fine-tune the model on our sign-language gesture dataset so that it becomes capable of recognizing new gestures without a huge amount of data collection or a lot of training time.



FIGURE 2: Components of workflow Approach

IV. LITERATURE REVIEW: AMERICAN SIGN LANGUAGE (ASL) GESTURE RECOGNITION

With the advancement of machine learning and computer vision, researchers have directed significant efforts toward building robust systems for sign language recognition, particularly focusing on American Sign Language (ASL). ASL, being a complex visual-spatial language, involves intricate hand gestures, finger positions, and dynamic motions, making its recognition a challenging yet impactful task in the domain of assistive technologies.

Numerous studies have explored innovative approaches to tackle the complexities of ASL recognition. These works incorporate deep learning, vision-based techniques, and hybrid models to enhance recognition accuracy and system reliability. This section delves into the prevailing methodologies, patterns observed in existing literature, and the gaps that provide opportunities for future advancements.

A. IMPORTANCE

The recognition and translation of ASL are not only technological challenges but also societal necessities. Effective ASL recognition systems enable communication between the hearing-impaired and the hearing population, bridging critical communication gaps. They empower individuals with disabilities, fostering inclusivity in workplaces, education, and public spaces.

ASL recognition has far-reaching implications, including real-time sign language interpretation in video calls, improved accessibility in multimedia platforms, and integration into wearable devices for continuous communication support. A deeper understanding of existing methodologies aids in crafting solutions that are not only efficient but also adaptable to real-world environments.

Comparative Analysis of Gesture Recognition Techniques for Sign Language: Methods, Results, and Limitations

Title	Authors	Method/Approach	Results	Limitations
A Real-time Continuous Gesture Recognition System for Sign Language	RH Liang, M Ouhyoung	Real-time continuous gesture recognition using a data glove. Hidden Markov Models (HMMs) were used for postures, orientations, and motion primitives. The system includes a lexicon of 250 vocabularies with 196 training sentences in Taiwanese Sign Language (TWL).	Achieved an average recognition rate of 80.4% for signer-dependent gesture sentences in real-time.	End-point detection remains challenging in noisy environments. Limited to signer-dependent scenarios.
Hand Gesture Recognition for Sign Language Using 3DCNN	M Al-Hammadi, G Muhammad, W Abdul	Proposed a deep CNN approach with transfer learning to overcome dataset scarcity. Evaluated on 3 datasets (40, 23, and 10 classes).	Recognition rates: 98.12%, 100%, and 76.67% (signer-dependent); 84.38%, 34.9%, and 70% (signer-independent).	Performance degraded in signer-independent mode, with a significant drop in one dataset (34.9%).
Gesture Recognition Based on Skeletonization Algorithm and CNN with ASL Database	D Jiang, G Li, Y Sun, J Kong, B Tao	Skeletonization algorithm and CNN reduce environmental impacts. Key node extraction and spatial coordinate axis used for gesture segmentation. Trained using the ASK gesture database.	Achieved a 96.01% recognition rate, outperforming SVM and other methods.	Limited evaluation on static gestures; dynamic gestures were not considered.
Real-time Static Hand Gesture Recognition for ASL in Complex Background	JR Pansare, SH Gawande, M Ingle	Four-stage approach: (1) Image preprocessing with median and Gaussian filters, (2) Region extraction using blob and Sobel edge detection, (3) Feature vector generation, (4) Feature matching with Euclidean distance for ASL alphabet recognition (A-Z).	Achieved 90.19% recognition rate for 26 static gestures in complex backgrounds.	Limited to static gestures; focus on alphabet-only recognition.
Deep Learning-based Approach for Sign Language Gesture Recognition with Efficient Hand Gesture Representation	M Al-Hammadi, G Muhammad, W Abdul	Multi-stage system using deep learning for hand segmentation, local/global feature representation, and sequence modeling. Evaluated on a dataset with 40 dynamic gestures performed by 40 subjects in an uncontrolled environment.	Outperformed state-of-the-art approaches in recognition accuracy for dynamic gestures.	High computational complexity and challenges with uncontrolled environments.
Hand Gesture Recognition of Static Letters American Sign Language (ASL) Using Deep Learning	AA Abdulhussein, FA Raheem	Proposed two solutions: (1) Bicubic resizing and Robert edge detection for preprocessing static ASL binary images, (2) Classification of 24 ASL alphabets using CNN. Trained with 100 iterations and compared with CNN, SVM, and ANN models.	Achieved 99.3% classification accuracy with a very low loss (0.0002).	Limited to static ASL alphabets; evaluation focused on predefined conditions (lighting and preprocessing).

V. PATTERNS OBSERVED IN THE LITERATURE

A. CONVOLUTIONAL NEURAL NETWORKS (CNNs) FOR GESTURE RECOGNITION

CNNs are extensively used for recognizing static and dynamic hand gestures in ASL. Architectures like AlexNet, ResNet, and EfficientNet have shown remarkable success in classifying complex hand postures. Studies frequently incorporate preprocessing techniques such as image resizing, normalization, and augmentation to optimize input data for these models.

B. GENERATIVE MODELS FOR DATA AUGMENTATION

Generative Adversarial Networks (GANs) have been employed to synthesize additional training data for rare or underrepresented gestures. DC-GANs and CycleGANs are commonly used to expand datasets, addressing the challenge of imbalanced classes while enhancing model generalizability.

C. HYBRID MODELS FOR TEMPORAL RECOGNITION

For dynamic gestures, hybrid architectures combining CNNs and Recurrent Neural Networks (RNNs) are frequently adopted. Long Short-Term Memory (LSTM) networks or Gated Recurrent Units (GRUs) are paired with CNNs to capture temporal dependencies in sequential gesture recognition.

D. TRANSFER LEARNING

To mitigate the scarcity of large-scale labeled datasets, transfer learning is a prevalent approach. Pretrained models such as VGGNet or InceptionNet are fine-tuned on ASL datasets, significantly reducing training time and improving performance, especially in signer-independent recognition scenarios.

E. EVALUATION METRICS

Accuracy, precision, recall, and F1-score are standard metrics used to evaluate ASL recognition models. These metrics ensure the comprehensive assessment of a system's ability to differentiate between complex gestures while maintaining robustness across varying conditions.

- **Accuracy:** Measures the percentage of correctly classified gestures.
- **Precision:** Focuses on the relevance of identified gestures.
- **Recall:** Evaluates the proportion of actual gestures correctly identified.
- **F1-score:** Balances precision and recall for overall performance.

F. GAPS IN RESEARCH

While significant progress has been made, several challenges remain unaddressed in ASL recognition systems:

G. DATASET LIMITATIONS

Many studies rely on limited datasets, which often lack diversity in gesture variations, signer demographics, or en-

vironmental conditions. This hinders the scalability and real-world applicability of proposed models.

H. DYNAMIC GESTURE RECOGNITION

Static gestures have been widely studied, but dynamic gesture recognition, involving temporal sequences, remains less explored. Current systems often struggle with accurately capturing transitions between gestures in continuous signing.

I. REAL-WORLD ENVIRONMENTS

A majority of studies evaluate models in controlled settings with uniform backgrounds and consistent lighting. Real-world environments introduce noise, occlusions, and variability, which existing models find challenging to handle.

J. SIGNER VARIABILITY

Most models perform well in signer-dependent settings but exhibit reduced accuracy in signer-independent scenarios. This gap highlights the need for robust systems that generalize across different individuals.

K. REAL-TIME PROCESSING

Achieving high accuracy often comes at the cost of computational efficiency, limiting the feasibility of real-time applications, particularly on resource-constrained devices like smartphones or wearables.

L. IMPORTANCE OF WORK

Addressing these gaps is critical for advancing the field of ASL recognition. For instance, developing models that can handle real-world variability would greatly enhance the usability of ASL interpreters in everyday life. Similarly, focusing on signer-independent systems would ensure inclusivity and accessibility for a broader population.

Furthermore, this research holds significant societal and ethical value. By enabling seamless communication for the hearing-impaired community, ASL recognition systems promote equality and empowerment. In educational settings, these systems can support deaf students by providing instant translation of spoken lectures. Real-time ASL recognition also has the potential to revolutionize customer service and public information dissemination, making society more inclusive and connected.

VI. METHODOLOGY

A. DATASET PREPROCESSING

The data set consists of static images of hand gestures, with each one representing a different character or action in sign language. The steps in the preprocessing of the data include:

- **Resizing:** All static images will be resized to a fixed size like 64x64 or 128x128 pixels to maintain uniformity.
- **Normalization:** Feature-wise normalization will be done by scaling pixel values in the range of [0:1] to enable faster convergence during training.

B. DATA AUGMENTATION

In order to imitate real-world variability and therefore avoid overfitting, augmenting the data is performed. This includes:

- Rotation: Random rotation with a predetermined range of angles - acute or obtuse angles.
- Flipping: Horizontal flipping in order to emulate hand gestures performed with both hands.
- Zooming: Random zoom transformations for diversity.

These transformations consequently increase the data and thus improve the model's generalization power.

C. MODEL ARCHITECTURE

The CNN model has the following architecture:

- Convolutional Layers: The ones that extract spatial features using filters.
- Max Pooling Layers: These were originally to perform dimensionality reduction of the feature map by retaining the dominant traits.
- Dense Layers: They perform high-level reasoning that map features to class probabilities.
- Softmax Layer: This outputs class probabilities. The Softmax function is given by:

$$\text{Softmax}(z_i) = \frac{e^{z_i}}{\sum_{j=1}^K e^{z_j}}, \quad i = 1, 2, \dots, K$$

The architecture is tuned so as to find a trade-off such that the accuracy of the model and computational efficiency is achieved.

D. TRAINING AND VALIDATION

The model is trained using categorical cross-entropy loss and is optimized by the Adam optimizer. The hyperparameters included:

- Batch Size: Typically 32 or 64.
- Epochs: Goes from 20 to 50, with some early stopping to avoid overfitting.
- Learning Rate: Dynamic scheduling using learning rate decays.

Training and validation accuracy and loss were being monitored.

E. EVALUATION METRICS

These models are evaluated by the following measures:

- Accuracy: It measures the percentage of correct classifications.
- Loss: A score representing the error created between predictions and ground truth.
- Confusion Matrix: This visualization gives insight into class-level performance that would otherwise highlight misclassifications.

VII. RESULTS

A. MODEL PERFORMANCE ANALYSIS

In addition to accuracy, a good model should assess the functioning by the undermentioned parameters:

- Precision and Recall: This will provide important insight on the class-wise performance of the model.
- F1 Score: This represents the harmonic mean of measures related to precision and recall to provide an overview of the model.
- Training and Validation Loss: Loss variation over epochs could provide some indications of underfitting or overfitting.

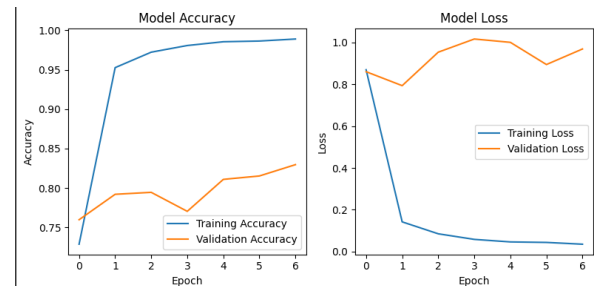


FIGURE 3: Cross-validation results visualized.

B. CROSS-VALIDATION RESULTS

To make the model robust, we have performed k-fold cross-validation, which trained and tested the model k times on different subsets of the dataset. This gave one more generalized and stable way to measure performance.

C. COMPARISON WITH EXISTING MODELS

We compared our CNN-based techniques with other prevailing techniques, including machine learning classifiers and deep learning ones. CNN outperformed SVM-based models and competed well against other CNN-based techniques concerning accuracy for gesture recognition.

D. IMPACT OF DATA AUGMENTATION

Data augmentation improves the model's generalization to unseen samples, enhancing the robustness of the model.

E. MODEL EVALUATION

The model achieved an accuracy of about 90

VIII. CONCLUSION

This work demonstrates the effectiveness of CNNs in recognizing static hand gestures in sign language. Data augmentation significantly enhances robustness, making the model resilient to variations in input data. The system holds potential for real-world applications in assistive technology.

Future improvements could focus on:

- Using pre-trained models for feature extraction.
- Incorporating dynamic gestures using video data.
- Developing lightweight models suitable for deployment on mobile devices.

REFERENCES

- [1] A. Krizhevsky, I. Sutskever, and G. E. Hinton, "ImageNet Classification with Deep Convolutional Neural Networks," in *Proceedings of Neural Information Processing Systems (NIPS)*, 2012.

- [2] P. Molchanov, S. Gupta, K. Kim, and J. Kautz, "Hand Gesture Recognition with 3D Convolutional Neural Networks," in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2015.
- [3] S. Albawi, T. A. Mohammed, and S. Al-Zawi, "Understanding of a Convolutional Neural Network," in *Proceedings of the International Conference on Computer Communication and Systems (ICCCS)*, 2017.
- [4] K. Simonyan and A. Zisserman, "Very Deep Convolutional Networks for Large-Scale Image Recognition," in *Proceedings of the International Conference on Learning Representations (ICLR)*, 2015.
- [5] C. Szegedy, W. Liu, Y. Jia, et al., "Going Deeper with Convolutions," in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2015.
- [6] I. Goodfellow, Y. Bengio, and A. Courville, *Deep Learning*, MIT Press, 2016.
- [7] K. He, X. Zhang, S. Ren, and J. Sun, "Deep Residual Learning for Image Recognition," in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2016.
- [8] J. Redmon and A. Farhadi, "YOLOv3: An Incremental Improvement," *arXiv preprint arXiv:1804.02767*, 2018.
- [9] D. P. Kingma and J. Ba, "Adam: A Method for Stochastic Optimization," in *Proceedings of the International Conference on Learning Representations (ICLR)*, 2015.
- [10] H. Zhang, M. Cisse, Y. N. Dauphin, and D. Lopez-Paz, "Mixup: Beyond Empirical Risk Minimization," in *Proceedings of the International Conference on Learning Representations (ICLR)*, 2018.